

Financial Econometrics I

Modeling Volatility

(Lecture 4)

Summer Semester 2025/2026 by Lukas Vacha

* If viewed in .pdf format - for full functionality use *Mathematica* 7 (or higher) notebook (.nb) version of this .pdf

Characteristics of Volatility

Volatility is one of the most important concepts in finance, as it measures the risk of financial assets. Volatility is not directly observable, it can be just estimated. It has some important characteristics:

- volatility clusters - After a large (small) price change a large (small) price change tends to occur.
- volatility does not diverge to infinity - it varies within some fixed range.
- negative and positive price shocks tend to have different impacts on volatility (leverage effect)
- heavy-tailed, non-Gaussian distribution

Historical Volatility

Most straightforward way to measure volatility is to estimate time-series of variance on "rolling samples" For zero-mean variable, this would be:

$$\sigma_t^2 = (r_{t-1}^2 + r_{t-2}^2 + \dots + r_{t-q}^2) / q,$$

where q is the latest observation used. This method may produce abrupt changes in estimates.

Exponential volatility

Alternatively, exponential moving average (EMA) estimator of volatility can be used. It uses all data points, and recent observations carry larger weights. (weights are exponentially decreasing):

$$\sigma_t^2 = \frac{\sum_{\tau=1}^m \lambda^{\tau-1} r_{t-1-\tau}^2}{1 + \lambda + \lambda^1 + \dots + \lambda^{m-1}},$$

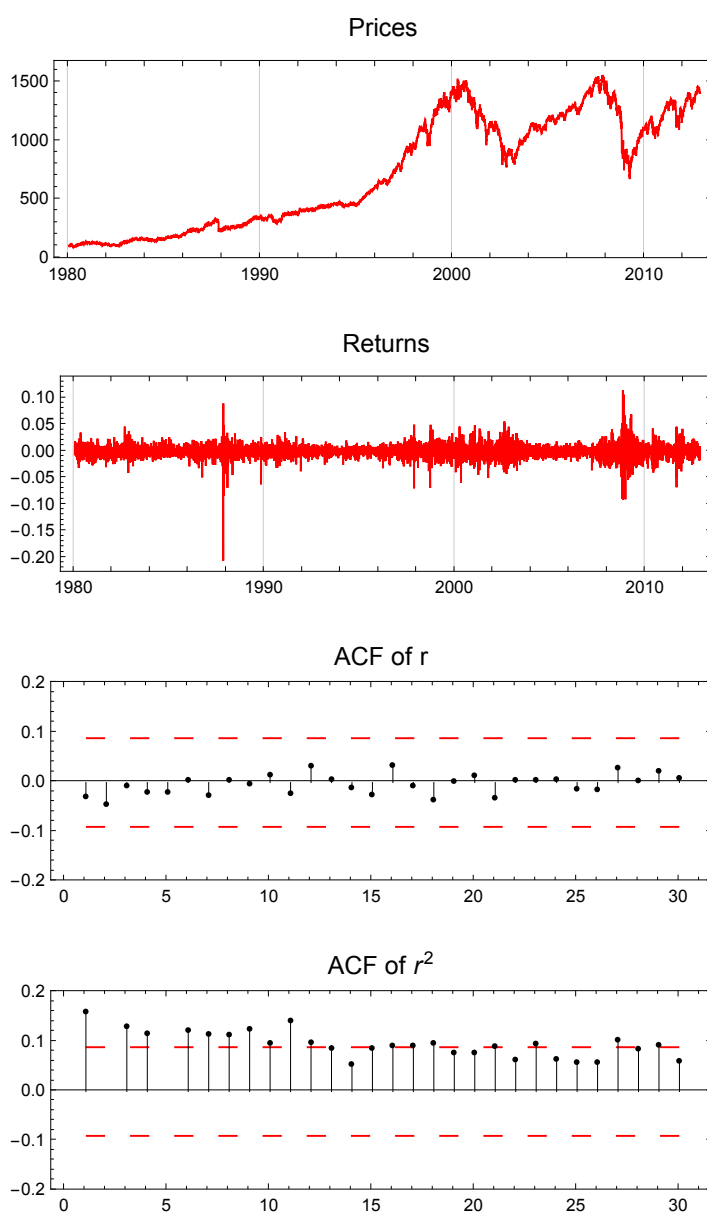
finally we get (we derive it next week):

$$\sigma_t^2 = (1 - \lambda)r_{t-1}^2 + \lambda\sigma_{t-1}^2,$$

where initial value could be unconditional variance in a historical sample. In most financial applications, $\lambda=0.94$ is used.

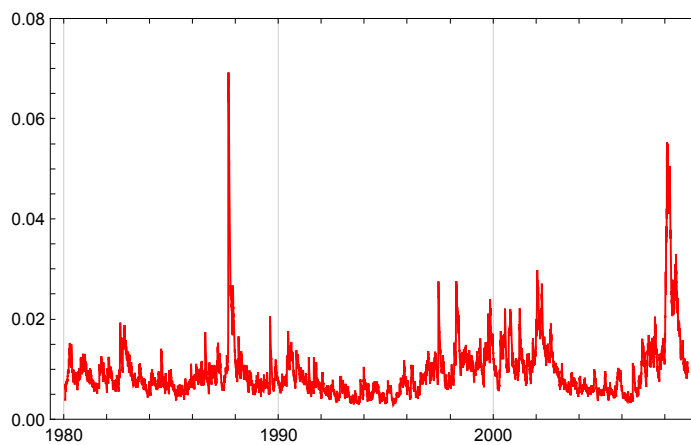
Example on S&P 500 index data

Note that characteristics of volatility can be observed from returns of S&P 500. Moreover, ACF plot suggests us, that returns are not serially correlated, but ACF plot of squared returns shows, that they are not independent. By modeling volatility, we will attempt to capture this kind of dependence.



Conditional Standard Deviation estimated by EMA (or EWMA)

S&P standard deviation estimate with EMA ($\lambda=0.9$)



Conditional Heteroscedastic Models

Log returns of an asset at time is either serially uncorrelated, or with minor lower order serial correlations, but it is **dependent**. Conditional heteroscedastic models are used for modeling σ_t^2 as volatility estimate by "allowing" heteroscedasticity (time-variation) and capturing that dependence.

ARCH(1)

The first model that provides a systematic framework for volatility modeling is Autoregressive conditional heteroscedasticity (ARCH) model. ARCH(1) model is most common from ARCH(m) models in financial theory. Basic idea is that the mean-corrected return is serially uncorrelated, but dependent, and that the dependence can be described by a simple function of lagged values:

$$\begin{aligned} r_t &= \mu_t + a_t \\ a_t &= \sigma_t \epsilon_t \\ \sigma_t^2 &= \alpha_0 + \alpha_1 a_{t-1}^2 \end{aligned}$$

Where a_t is mean corrected return and $\alpha_0 > 0$, $\alpha_1 \geq 0$.

Unconditional mean of a_t is zero, unconditional variance is $\alpha_0 / (1 - \alpha_1)$.

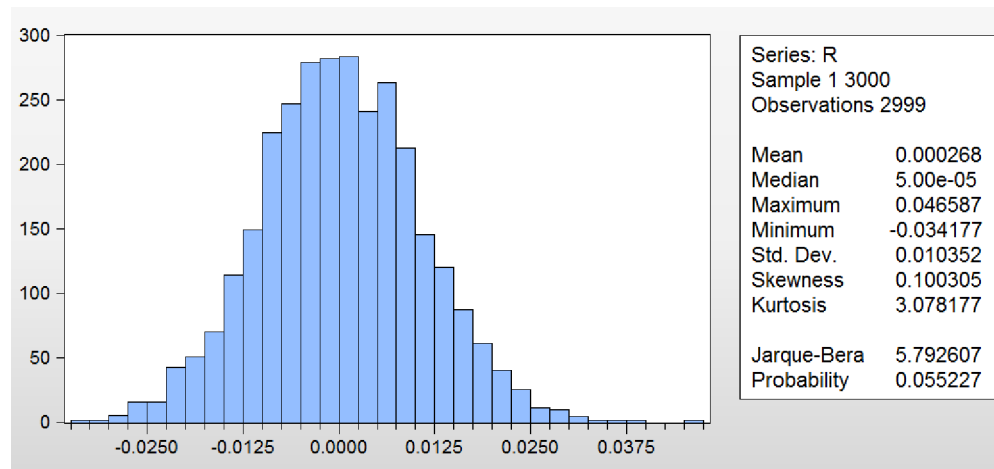
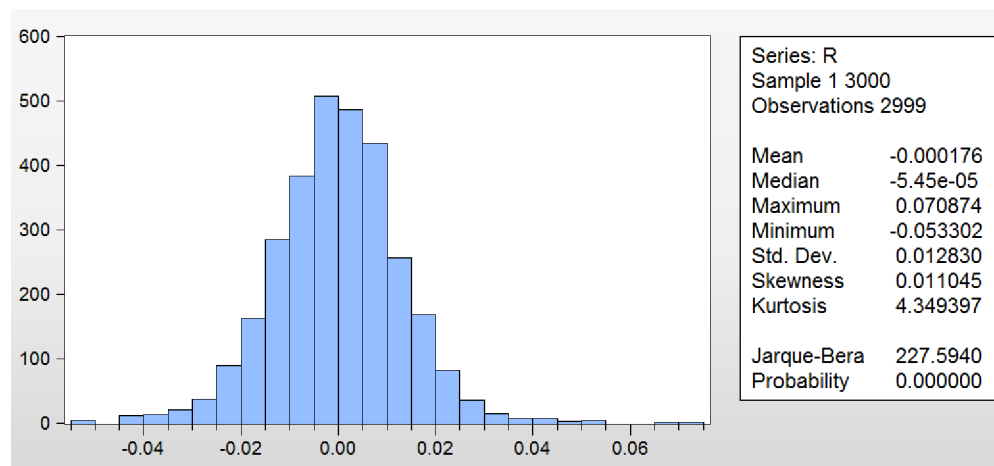
The process is stationary, if $\alpha_1 < 1$, the fourth moment is finite if $3\alpha_1^2 < 1$.

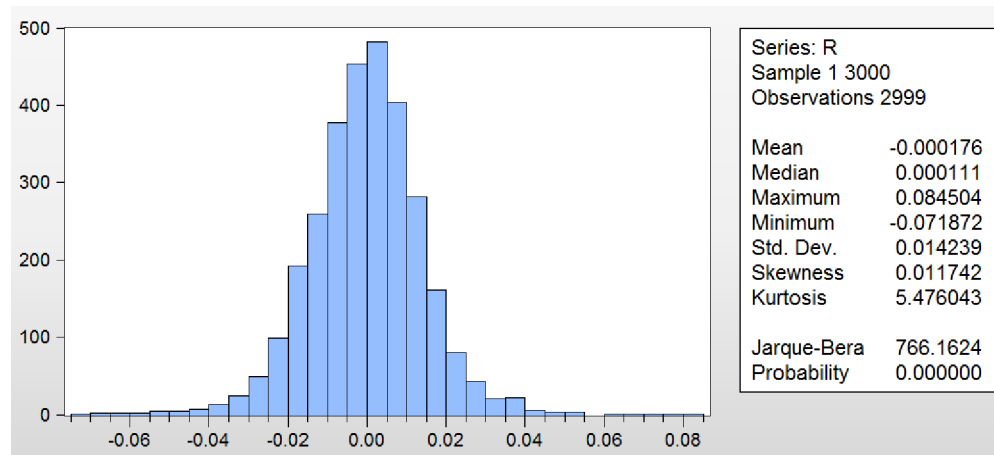
Kurtosis of ARCH(1) processes

The fourth moment of ARCH(1) process:

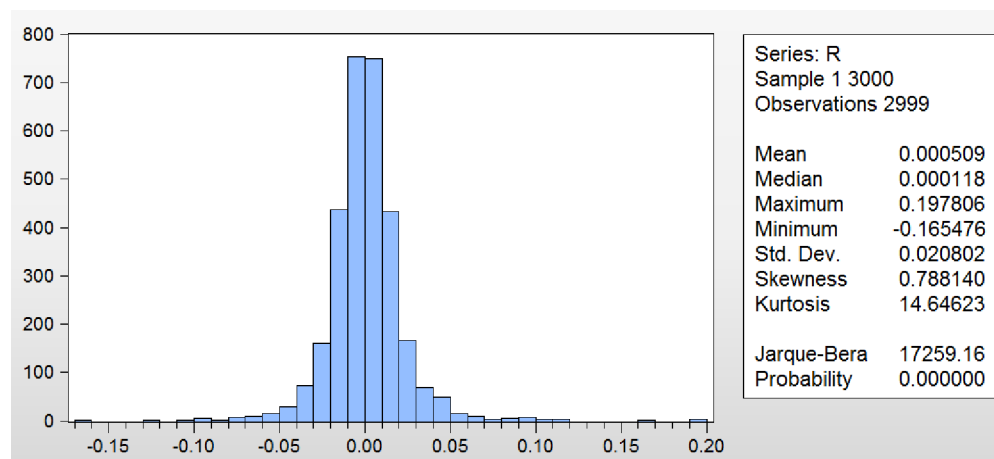
$$E[\epsilon_t^4] = \frac{3\alpha_0^2}{(1-\alpha_1)^2} \frac{1-\alpha_1^2}{1-3\alpha_1^2}, \text{ with } 3\alpha_1^2 < 1$$

$$\text{kurt}(\epsilon_t) = \frac{E[\epsilon_t^4]}{E[\epsilon_t^2]^2} = 3 \frac{1-\alpha_1^2}{1-3\alpha_1^2} \geq 3$$

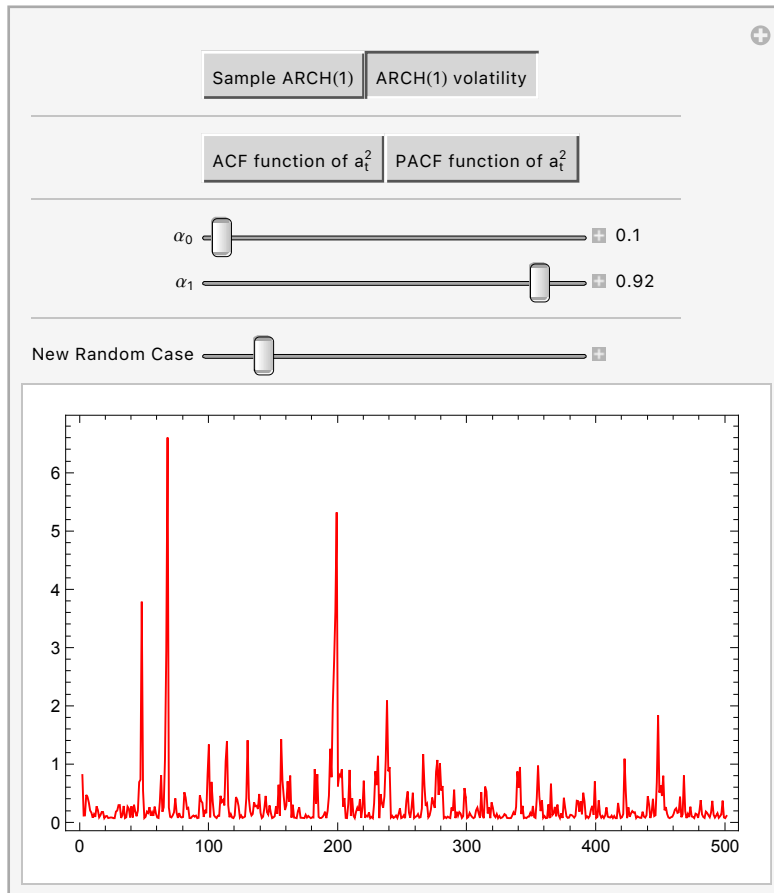
Example : $\alpha_1 = 0.1$ **Example : $\alpha_1 = 0.4$** **Example : $\alpha_1 = 0.5$**



Example : $\alpha_1 = 0.9$



Examples of ARCH(1) artificial processes



ARCH(m)

$$a_t = \sigma_t \epsilon_t,$$

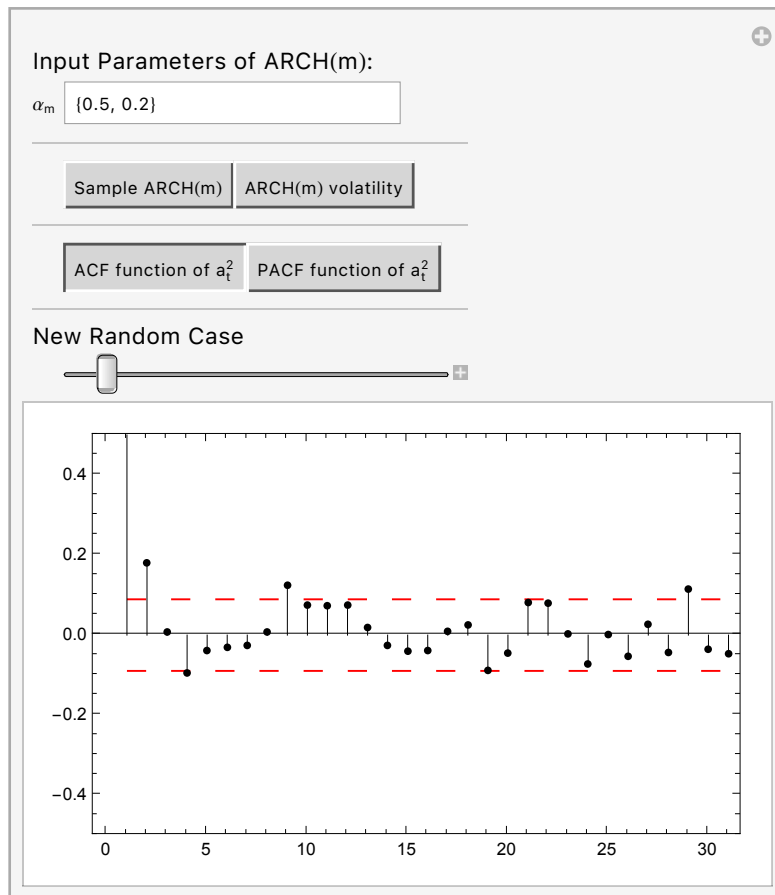
$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \dots + \alpha_m a_{t-m}^2,$$

where a_t is mean corrected return $a_t = r_t - \mu_t$, $\{\epsilon_t\}$ is i.i.d random variables with zero mean and variance 1, $\alpha_0 > 0$, $\alpha_i \geq 0$ for $i > 0$.

One can easily observe, that large past squared shocks $\{a_{t-i}^2\}_{i=1}^m$ imply large conditional variance σ_t^2 , or volatility. Thus under ARCH, large shocks tend to be followed by large shocks - ARCH effect

Examples of ARCH(m) artificial processes

Note that you have to input $\alpha_0 > 0$, $\alpha_i \geq 0$ for $i > 0$. in form $\{\alpha_0, \alpha_1, \dots, \alpha_m\}$



Test for ARCH effects in time series

We use Lagrange Multiplier test (LM test), with the null hypothesis of no ARCH effects:

$$\epsilon_t^2 = \alpha_0 + (\sum_{s=1}^q \alpha_s \epsilon_{t-s}^2) + v_t$$

$$H_0 : \alpha_1 = \dots = \alpha_s = 0$$

$$LM = N.R^2 \sim \chi^2(p)$$

Forecasting with ARCH

Forecasts of the ARCH can be obtained recursively at origin h , the 1-step ahead forecast of σ_{h+1}^2 is:

$$\sigma_h^2(1) = \alpha_0 + \alpha_1 \epsilon_h^2 + \dots + \alpha_m \epsilon_{h+1-m}^2$$

l -step ahead forecast will be:

$$\sigma_h^2(l) = \alpha_0 + \sum_{i=1}^m \alpha_i \sigma_h^2(l-i),$$

where $\sigma_h^2(l-i) = \epsilon_{h+l-i}^2$ if $l-i \leq 0$.

Weaknesses of ARCH Models

ARCH model has also following weaknesses:

- assumes that positive and negative shocks have the same effect on volatility, but empirical testing shows, that assets respond differently to positive and negative shocks.
- model gives us description of the conditional variance, but does not explain the real behavior and source of variance.
- ARCH models tend to over-predict the volatility as they respond slowly to large isolated shocks.

GARCH models

GARCH(1,1)

GARCH(1,1) is most common for financial applications, as the dependencies are mostly very weak. It takes form of:

$$a_t = \sigma_t \epsilon_t,$$

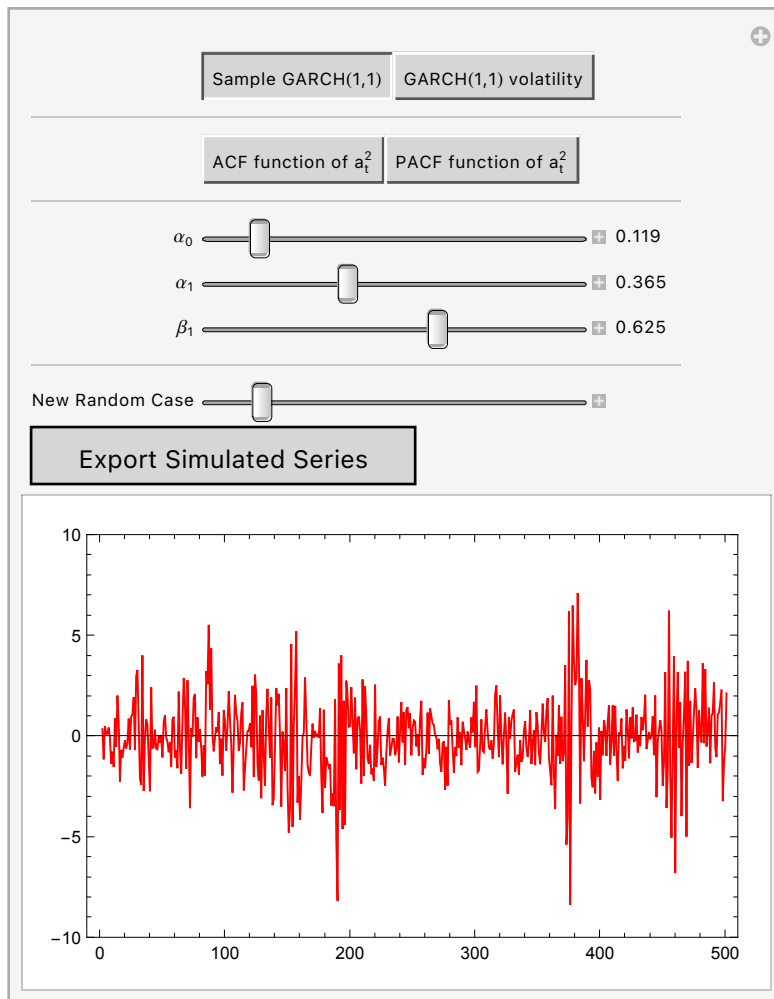
$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where $(\alpha_1 + \beta_1) < 1$. Large past shocks of return and volatility in $t-1$ gives large shocks to volatility in time t , thus volatility clustering is quite well captured.

Example GARCH(1,1) artificial processes

Note that Condition $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$ must be met.

Initial values are set to simulate GARCH(1,1) process with $\alpha_0 = 0.5$, $\alpha_1 = 0.2$, $\beta_1 = 0.4$



GARCH(m, s)

As ARCH model requires many parameters to adequately describe volatility process of returns, extension is used - Generalized autoregressive conditional heteroscedasticity (GARCH) model. It is assumed, that mean equation is adequately described by ARMA, letting $a_t = r_t - \mu_t$ be the mean-corrected log return, GARCH(m, s) model is described by following equation:

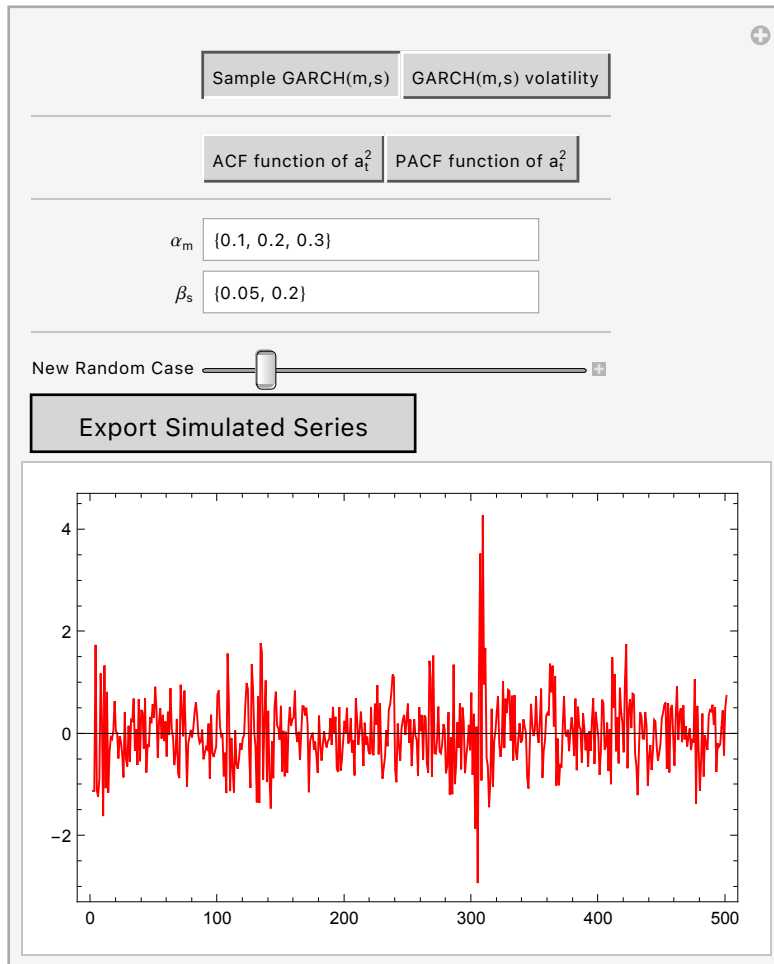
$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2,$$

where $\{\epsilon_t\}$ is i.i.d random variables with zero mean and variance 1, $\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$, so unconditional variance of a_t is finite, whereas its conditional variance σ_t^2 evolves over time.

Note, that GARCH model reduces to a pure ARCH(m) model for $s = 0$.

Example GARCH(m,s) artificial processes



GARCH-M model

The return of a security may sometimes depend directly on volatility. To model this, we use GARCH in mean (GARCH-M) model.

GARCH(1,1) - M is formalized as:

$$r_t = \mu + c\sigma_t^2 + a_t$$

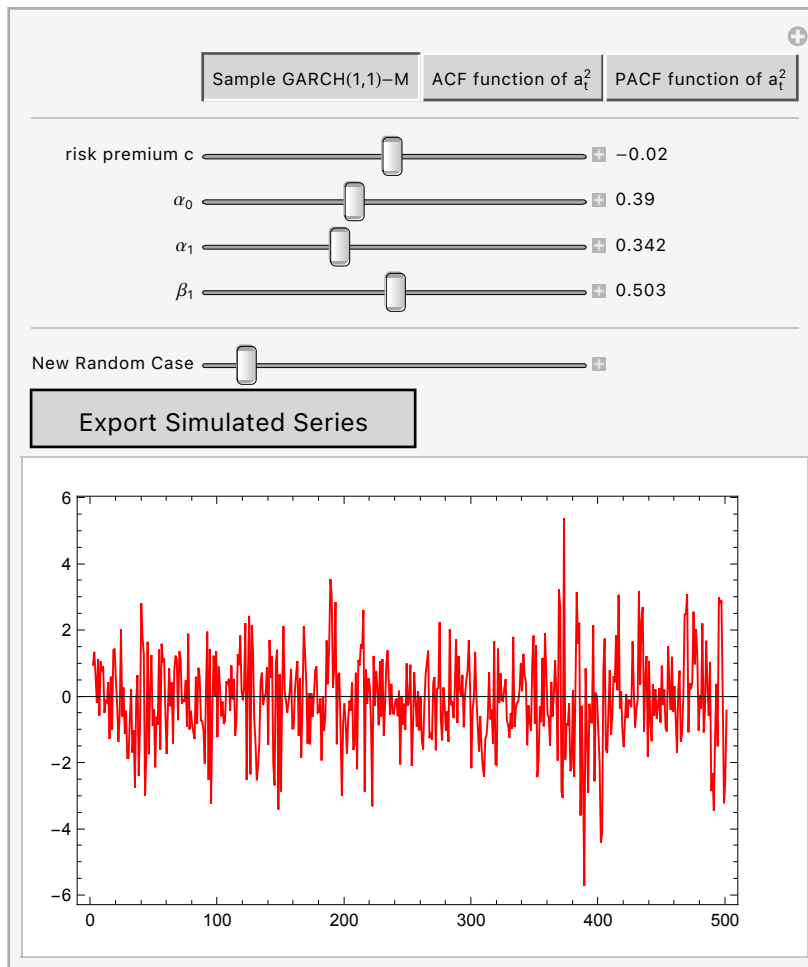
$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where μ and c are constant. c is also called risk premium

Example GARCH(1,1)-M artificial processes

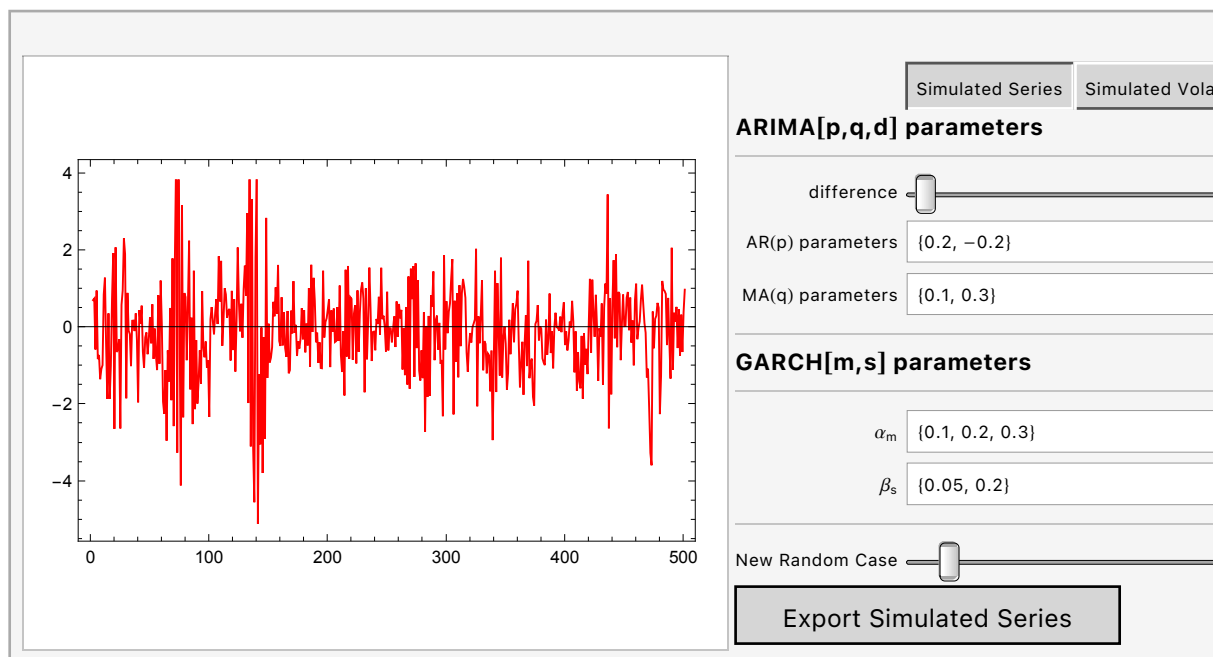
Note, that with positive risk premium c , returns are positively skewed, as they are positively related to its past volatility



ARIMA-GARCH Models

Common way to build ARCH model is to remove any linear dependencies in the data (i.e., by ARMA model). For most series, we remove also sample mean from the data, and use residuals for testing the ARCH effects. This can be done either using Ljung-Box statistics, or ACF, PACF functions, or by Lagrange multiplier test (LM test). If the statistic is significant, then conditional heteroscedasticity of a_t is detected, and ACF of a_t^2 is used to determine order of ARCH effect. For GARCH, it is not so easy to determine the order, but empirical findings can serve again, that in most financial applications, we use lower order models such as GARCH(1,1), GARCH(2,1).

Example ARIMA(p,d,q)-GARCH(m,s) artificial processes



⋮

IGARCH - Integrated GARCH

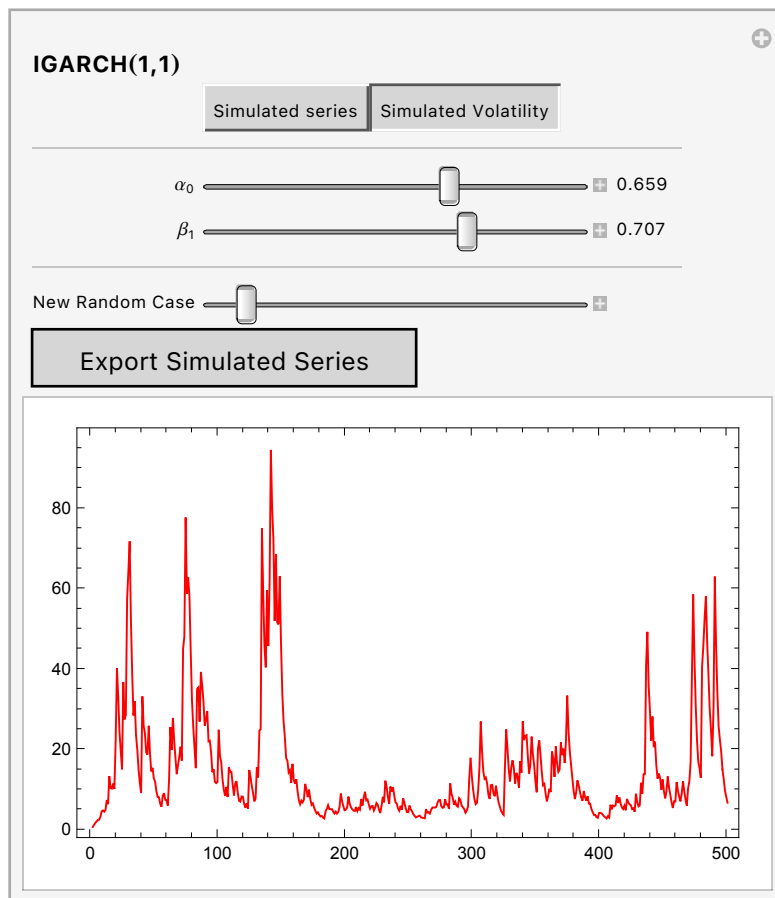
IGARCH models are unit-root (integrated) GARCH models, their key feature is, that past squared shocks is persistent.

IGARCH(1,1) is formalized as:

$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) a_{t-1}^2$$

Example IGARCH(1,1) artificial processes



Reading: Starica, C. "Is GARCH (1, 1) as good a model as the Nobel prize accolades would imply." Preprint (2003). <https://notendur.hi.is/~helgito/NobelGarch.pdf>